Attributing annotator polarization to sociodemographic groups

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Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

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1 Introduction

Annotations are essential for a wide range of tasks, especially in fields such as content moderation, toxicity detection and hate speech detection. These tasks are especially essential for training systems that protect vulnerable and minority groups in online spaces, where they are often targeted []. However, annotations for such tasks are subjective and usually based on majority-group annotators. This may at best limit the usefulness of the data, or at worst lead to biased and misconfigured systems.

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree in vain. An example of such cases would be racist comments that are presented with coded language (usually referred to as "dog-whistles" [1]), which are not picked up by most annotators, but only by ones belonging in the targeted minority. Inversely, majority-group annotators may bias their own annotations against the minority-group annotators. **(author?)** [2] for instance show that toxicity models disproportionally marked African American speech as “toxic”.

Inter-annotator agreement is often used to detect such cases [3]. Metrics based on agreement mostly measure annotation variance, which may not be an optimal problem formulation for detecting such cases. *Polarization* is a better instrument in this regard, since it formulates annotation analysis as a cluster identification problem [4, 5]—specifically whether two or more annotation clusters are present (Figure 2).

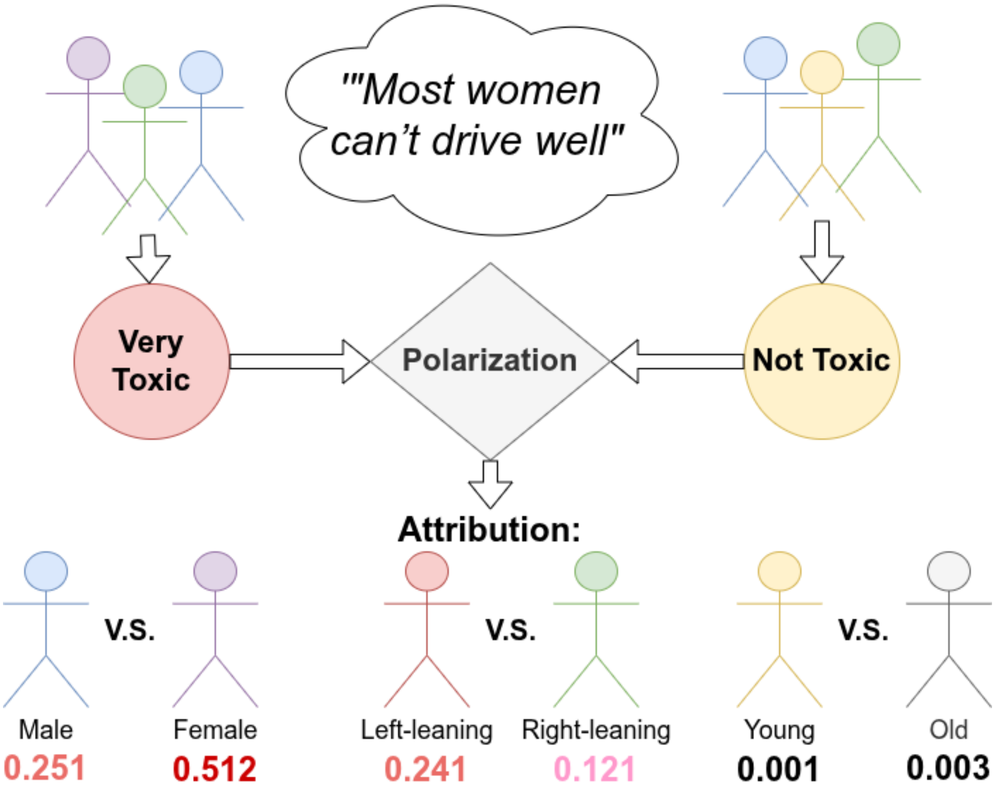

Fig. 1 The Apunim framework attributes annotator polarization to certain characteristics.

While we can detect polarization [4, 5], there is little work on how to detect whether it is actually caused by marginalized groups. **(author?)** [5] propose a mechanism which only works in the extreme case where overall polarization is zero; in this work we demonstrate that this case is not frequent in real-world data. We instead propose the *“Aposteriori Unimodality (Apunim) Metric”*, which attributes polarization to individual annotator sociodemographic groups. Our contributions are:

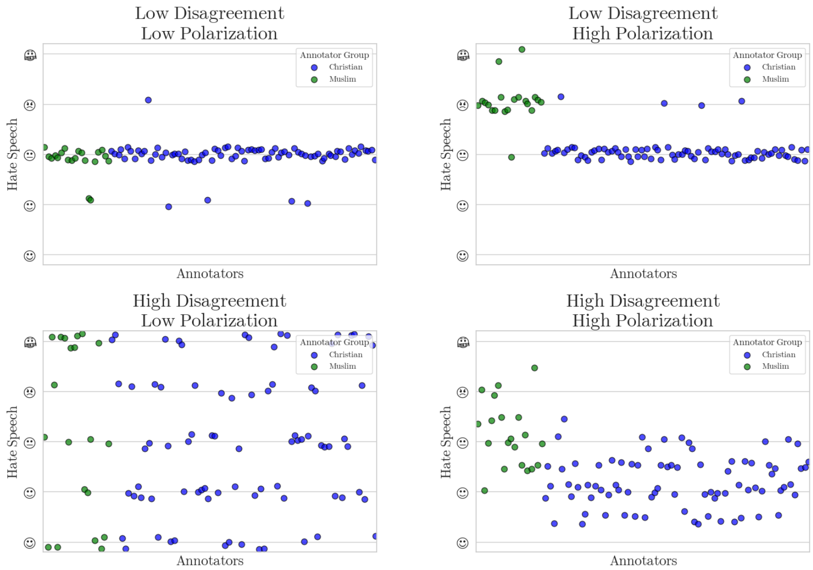

Fig. 2 Disagreement is similar to variance; thus it overlooks cases where annotators may be "split down the middle". Polarization on the other hand, is able to identify multiple clusters even if they are close together.

1. We introduce a new metric which measures how much polarization can be attributed specific sociodemographic groups.
2. We discover issues not yet acknowledged in literature, such as the presence of inherent polarization and conflicting polarization directions. Our proposed metric is designed to avoid these issues.
3. We apply our metric to four datasets; two with human comments and annotations [6, 7] as well as two generated and annotated by Large Language Model (LLM) agents [8].

2 Methodology

2.1 Problem Formulation

SocioDemographic Backgrounds

Since our goal is to pinpoint which specific characteristics contribute to polarization, we need to isolate individual groups within a SocioDemographic Background (SDB). Let Θ be one of multiple "dimensions" the set of all annotator SDB groups for a single dimension (e.g. "male", "female" and "non-binary" if we are investigating annotator gender).

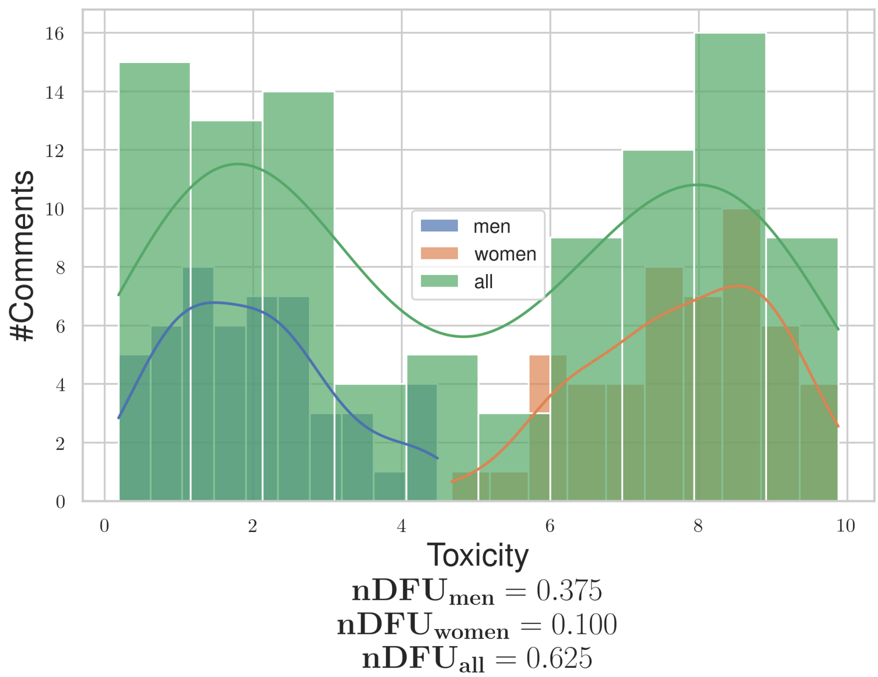

Fig. 3 Example of a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender.

Annotations

Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated data points. We assume that annotating a data point depends on two variables: inherent ("apriori") controversy, and the ("aposteriori") annotator's SDB, as well as uncontrolled factors such as mood and personal experiences (noise). Assuming that each data-point c is assigned multiple annotators, we can define the annotation multi-set $A(c) = \{(a, \theta)\}$, where a is a single annotation. As an example, the annotations for a comment in a sentiment analysis task would be formulated as:

$$\begin{aligned} &A(\text{"}\textit{Could be better, could be worse}\text{"}) = \\ &\{(positive, female), (negative, male), \\ &(positive, female), \ldots\} \end{aligned} \tag{1}$$

Polarization

Each set of annotations features a certain degree of *polarization*. Unpolarized annotations are usually unimodal; the annotators disagree on the details (which would be shown roughly as a bell curve around the median annotation), not fundamentally (which would likely be shown as a multimodal distribution). **(author?)** [5] create

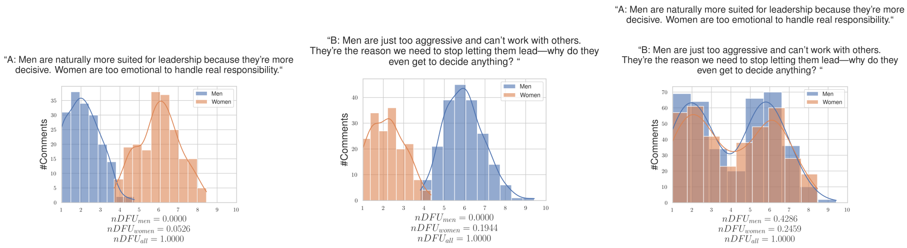

Fig. 4 Hypothetical example of a polarizing discussion with two comments, for which the annotators disagree on which is the toxic one. If we aggregate the two comments, the polarization scores for both men and women significantly rise, obscuring whether the exhibited polarization can be partially attributed to gender.

a polarization metric, the "normalized Distance From Unimodality (nDFU)", which measures whether an annotation set shows no polarization (unimodal distribution—$nDFU \approx 0$), up to complete polarization (multimodal distribution—$nDFU \approx 1$). Thus, we need a metric that measures how much the nDFU of a polarizing data point can be partially explained by θ.

2.2 Intuition

Comment polarization

Intuitively, θ partially explains the polarization of a data point c when the annotations grouped by θ are more polarized compared to the full set of annotations. Figure 3 exhibits a hypothetical example where a misogynistic comment is annotated for toxicity by male and female annotators.[1] The annotations are generally polarized ($nDFU_{all} = 0.625$). The annotations by female annotators exhibit low polarization between themselves ($nDFU_{women} = 0.1$), since most agree the data-point is toxic. The set of male annotations, also shows low polarization ($nDFU_{men} = 0.3725$), but for the opposite reason—most men agree that it is *not* toxic. This suggests that the overall polarization is driven by disagreements between male and female annotators.

Dataset polarization

Given this observation, we would be tempted to aggregate all annotations in the dataset and compare the polarization of each θ with the full set of annotations $A(c)$. In fact this is very tempting since aggregating annotations solves the very real and important problem of annotation scarcity; each data point may be annotated by only 5 annotators, leading to a best-case 3-2 split between groups. Since nDFU is at its core a histogram estimation, 3 annotations may not be enough to acquire a reliable histogram (this is discussed in depth in Appendix B).

However, this formulation would not work well. Figure 4 illustrates a hypothetical discussion with two comments, both of which are toxic, but where male and female annotators disagree on *which* comment is the toxic one. If we aggregate the annotations to a single discussion, the opposing polarization effects might balance each other

[1] The use of only two predominant genders is made only for the purposes of demonstration.

out, leading to a false negative. This example demonstrates that polarization has a *direction*, which is not captured by the (symmetric) $nDFU$ metric.

2.3 Aposteriori polarization

The observed polarization of a data-point on the group of annotations θ can be directly computed as:

$$pol(c, \theta) = nDFU(P(A(c), \theta) \tag{2}$$

where $P(A(c), \theta_i) = \{a(c;\theta) \in A(c) | \theta = \theta_i\}$ is the partition of A for a data-point c, for which its annotators belong to the SDB group θ_i.

In §1 we mentioned that annotator polarization can be caused either by the annotators SDB, or other factors. The latter can be considered "apriori" polarization, which is driven by the inherent "controversy" of the data point. Previous work [5] considered the case where this apriori polarization was zero; this is not the case in many datasets, as we demonstrate in §3.

Given infinite annotations, the assumption of independence would hold. Likewise, if we compared the subset with infinite random subsets of the superset. We can thus estimate the apriori polarization in data point c by bootstrapping. We randomly partition the annotations in $|\Theta|$ groups, with matching group sizes (e.g., if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we will create random partitions of sizes 80 and 20). The randomly partitioned annotations are then sampled t times. Formally, the estimated apriori polarization will be given by:

$$\frac{1}{t}\sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c))) \tag{3}$$

where $\tilde{P}_i$ is the random partition operator with matching group sizes.

While data scarcity is an important limitation when acquiring the observed polarization, this is decisively not the case with the number of data points themselves, which can range up to hundreds of thousands. Thus, we have the luxury of filtering data points before computing a dataset-wide metric. Since our metric attributes polarization, unpolarized data points are useless. The same applies for data points which are composed of only a single annotator group. Thus, we define the subset S_d of dataset d such that:

$$\begin{aligned} S_d = \{c \in d \mid\ & nDFU(c) > \alpha \\ & \textit{and c annotated by at least two groups}\} \end{aligned} \tag{4}$$

where α is a small positive number (in our experiments $\alpha = 0.2$).

By obtaining the pol-statistics for all data-points in S_d for SDB θ, we can get the mean observed polarization for the dataset:

$$pol_{obs.}(d, \theta) = \frac{1}{|d|}\sum_{c \in S_d} pol(c, \theta) \tag{5}$$

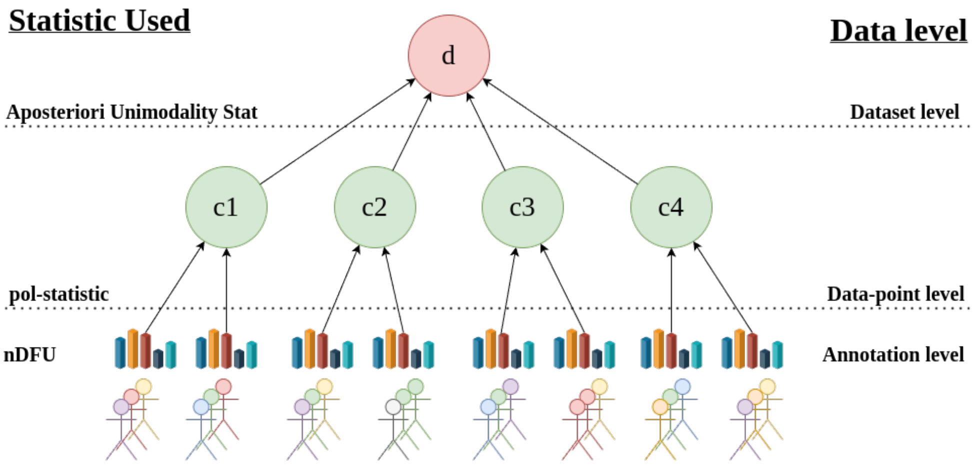

Fig. 5 An overview of the Aposteriori Unimodality Test. We gather statistical information from the annotations (different SDBs are denoted by different colors) via the nDFU measure, which is aggregated on the data-point-level by the pol-statistic. The Aposteriori Unimodality Test is applied on the discussion level, operating on the pol-statistics of the individual data-points.

and similarly the mean apriori dataset polarization:

$$pol_{apriori}(d) = \frac{1}{|d|} \sum_{c \in S_d} \frac{1}{t} \sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c))) \tag{6}$$

We can then define the *"Aposteriori Unimodality" (Apunim)* Statistic as:

$$apunim(d, \theta) = \frac{pol_{obs.}(d, \theta) - pol_{apriori}(d)}{1 - pol_{apriori}(d)} \tag{7}$$

Since $nDFU \to [0, 1] \Rightarrow apunim \to [-1, 1]$. If $apunim(d, \theta) \approx 0$, the exhibited polarization among annotators of group θ does not surpass what would be expected by chance. If $apunim(d, \theta) > 0$, then the group partially explains a rise in polarization, and inversely for the case $apunim(d, \theta) < 0$. The scope of the statistic and its relationship with the previously presented statistics are demonstrated in Figure 5.

2.4 p-value estimation

By obtaining the pol-statistics for all data-points in a dataset d we can apply any statistical means test with the null hypothesis:

$$H_0 : \forall \theta \in \Theta,\ apunim(d, \theta) \leq 0 \tag{8}$$

versus the (multiple) alternative hypotheses:

$$H_{\theta_i} : apunim(d, \theta_i) > 0 \tag{9}$$

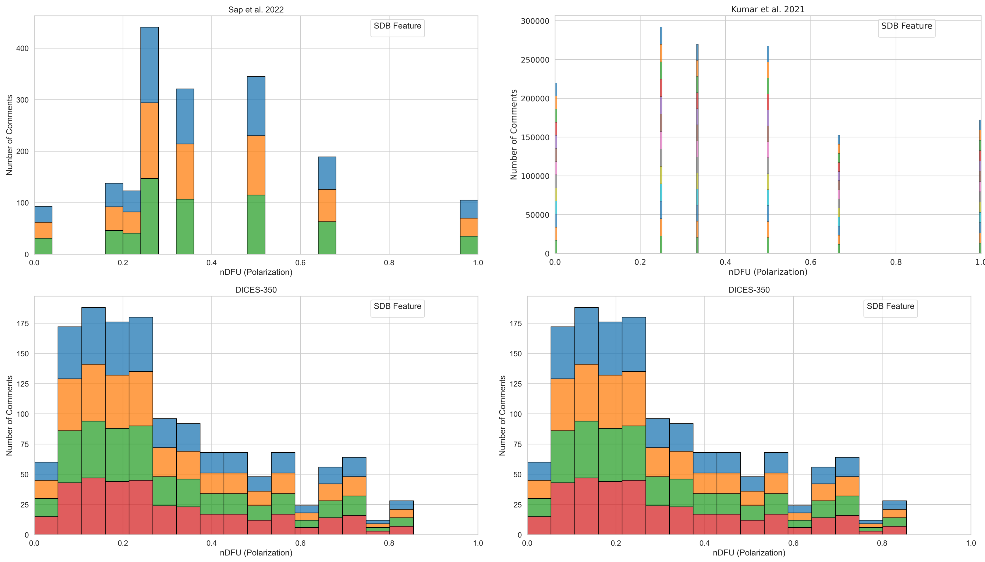

Fig. 6 Histogram of dataset polarization for the human datasets considered in this paper per SDB feature.

Our test assumes that (1) there exist enough data-points to reliably run the means-test, and (2) enough annotations in each data-point and SDB group to estimate the *pol-statistics*. We discuss the former assumption in Appendix B. The latter can be safely assumed for most datasets, because they feature enough data-points to reliably assume normal residuals ($n \gg 30$—see §3), which is why we use the Student-T test [9] .

Since we are simultaneously considering $|\Theta|$ hypotheses, we apply a multiple comparison correction to the resulting p-values. We choose the Holms method [10]. The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses [11]). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

3 Datasets

We use the datasets provided by **(author?)** [7] and **(author?)** [6]. They both feature various online comments, each annotated by a number of human annotators (5 and 4-6 annotators per comment respectively). The **(author?)** [7] dataset stands out for featuring more than 100,000 comments, while also including 10 different dimensions for SDBs and annotator experiences. We also use the DICES-350 and DICES-990 datasets [12], which are concerned with LLM response safety.[2] They feature less annotated comments, but more than an order of magnitude more annotators per comment,

[2]We specifically use the `Q_overall` annotations, which combine annotations for text that is considered harmful, (racially) biased and misinformation.

Table 1 Brief overview of the datasets used in this study. We are interested in the number of comments (#Comm.), annotators per comment (#Ann.) and number of SDB dimensions (#Dims.).

Name	# Comm.	# Ann.	# Dims
(author?) [7]	106,035	5-6	10
(author?) [6]	626	5-6	4
DICES-350 [12]	72,103	60-70	2
DICES-990 [12]	43,050	123	2

making them an invaluable resource for our study. While the datasets technically target different annotation criteria (racism, toxicity and LLM safety respectively), these criteria are closely related to each other.

Brief dataset characteristics are given in Table 1. There is a reasonable amount of polarization in most annotations for all datasets as demonstrated by Figure 6.

4 Results

4.1 Which factors contribute to polarization?

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided SDB dimensions can partially explain polarization in the toxicity/racism annotations. The results for the **(author?)** [7] and **(author?)** [6] datasets can be found in Tables **??** and **??** respectively, while Tables **??** and **??** display the results for the DICES-350 and DICES-990 datasets respectively.

We observe statistically significant polarization effects caused by annotator **Ethnicity** *on all datasets*

In the DICES dataset, non-white annotators seem to contribute to polarization much more than the majority white annotators. This effect is reversed for the **(author?)** [2] dataset.

Annotator **age** *does not cause polarization for toxicity/hate detection, but does for safety*

None of the age groups in the **(author?)** [2] and **(author?)** [7] datasets produce statistically significant results for polarization attribution, suggesting that it is not an important factor. However, both DICES datasets produced statistically and quantitatively significant results. This may be caused by the large number of annotators in these datasets, which enable our method to better distinguish this effect from noise in the annotations. This is unlikely however, given our next finding.

Polarization attribution follows ordinal ranking

While most SDB attributes are categorical, some are ordinal (e.g., Age, Education, LIKERT-scale opinions). We discover a pattern among (statistically significant) ordinal attributes, where polarization changes monotonically with their ranking. Examples include:

- Polarization increasing/decreasing with annotator age in both DICES datasets (but not in the **(author?)** [2] and **(author?)** [7] datasets, where there is no statistical significance).
- Thoughts on religion and toxicity in the **(author?)** [7] dataset (although much more noisily, since some attribute factors do not display statistical significance).

A notable exception is *Education* in the **(author?)** [7] dataset. This could be attributed to the curse of dimensionality, since we investigate a great number of groups, but there are few annotators for each individual group.

The fact that this behavior is not replicated when investigating annotator age in the **(author?)** [2] and **(author?)** [7] datasets lends further support to the argument that age does indeed not play a major role in annotator polarization for these datasets.

4.2 Investigating ordinal attributes

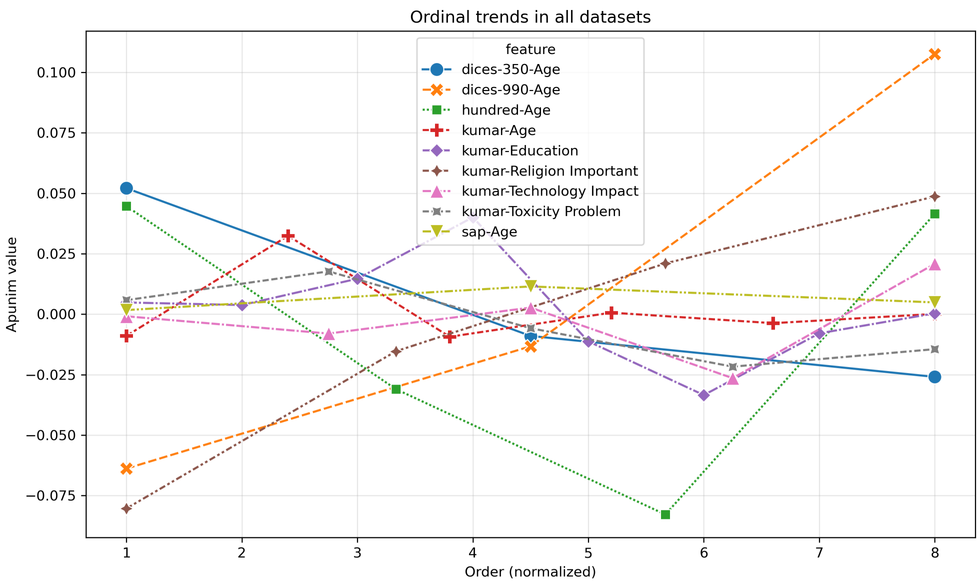

Fig. 7 The behavior of ordinal variables in all datasets. The x-axis represents the order of each factor (marked as `1), 2), ...` in Tables ??,??,??,??). The x-axis is normalized to the smallest common denominator between the number of ordinal values.

Most ordinal attributes (with statistically significant apunim values) tend to cluster around a few dominant values; for example, *Age* and *Education* in the **(author?)** [7] dataset. This pattern is expected because the Apunim algorithm is designed to identify clusters of annotations, causing nearby or similar values to be absorbed into the main effects driven by a few significant modes.

However, some ordinal attributes (*Age* in both DICES-350 and DICES-990, *Religion Importance* in **(author?)** [7]) follow a strict, monotonic trend. This implies that, in some cases, the **amount of polarization increases the further apart an opinion is from neutrality**. For example, non-religious annotators are less likely to have polarizing opinions about what constitutes a toxic comment than the average annotator, while very religious annotators generate very polarizing judgments.

5 Conclusion

We introduced the "Aposteriori Unimodality Test", a statistical test that detects whether certain sociodemographic groups partly cause polarization in annotations. Our test is resistant to low samples of annotations, bypasses common statistical issues such as sample dependence, and avoids newly-discovered issues such as the presence of inherent polarization and polarization direction. We apply our test two human-generated and two LLM-generated datasets and find interesting patterns on the former, and verify current literature on LLM SDB prompting on the latter.

6 Limitations

Since there are no other established tests that attribute polarization to sociodemographic characteristics, there is no way to verify that the results presented in §4 are accurate. Similarly, there is no qualitative way of evaluating our test, other than observations on the (non-) efficacy of LLM SDB prompting and our own intuition. Lastly, while our test seems to work sufficiently well with a small number of annotators per data-point, we still encourage researchers using this test to have at least a few annotators of each sociodemographic group that is being tested, in their dataset.

7 Ethical Considerations

While our work aims to help protect marginalized and disadvantaged groups by attributing polarization to certain subgroups, it can be taken advantage of by malicious actors. Given a dataset with fine-grained SDB information, these actors can use our test to target specific vulnerable groups. We thus urge researchers to keep datasets including such information protected, and provided to others only under explicit permission.

A Acronyms

LLM Large Language Model
SDB SocioDemographic Background
nDFU normalized Distance From Unimodality
FWER Family-Wise Error Rate

B How many annotators?

While our test works best with many annotations per datapoint, the cost required for multiple human annotators is often prohibitive [13]. It is thus very important both DICES datasets, as well as the **(author?)** [6] dataset annotated by 100 distinct LLM annotators.

The LLM annotation procedure is described in Appendix D. While a more appropriate comparison would entail synthetic annotations in the DICES datasets themselves, we elect to use a standard task for LLM-as-a-judge (hate speech detection) instead of the relatively challenging and under-explored area of general LLM safety.

B.1 Methodology

For each comment in all datasets, we randomly sample a subset of annotators and compute the apunim value. We repeat the procedure 30 times, and average all observations. This was a necessary step to reduce the variance inherent in random sampling. We also truncate the results for the DICES-350 dataset, since very few comments were annotated by the respective annotator counts.

B.2 Results

Figure 8 demonstrates the effect of the number of annotators on the reliability of the polarization estimation. We can see that the metric becomes consistent with about 20 annotators per comment, after which we start to get diminishing results. This experiment was conducted on a subset of the **(author?)** [6] dataset, with LLM annotators having different SDBs.

C Theoretical limitations

In §2.2 we compared the polarization of the whole set of annotations with the partitions by annotator group. This approach uses the same samples for both subset and superset, violating the fundamental hypothesis of independent samples. This also means the statistics for both will *always be similar to an extent*. This is especially pronounced in our case, due to the extreme overlap exhibited by tiny annotation sets. Additionally, when calculating p-values, the non-independence of samples induces an intrinsic, non-zero covariance between any statistic s calculated on A and on S.

A standard method of side-stepping this issue is to compare the subset with its complement to the superset. This can not be applied to our case. Consider the example in Figure 3; if we compared $nDFU_{men}$ with $nDFU_{women}$, the result would be very close to zero, indicating no polarization attribution to gender.

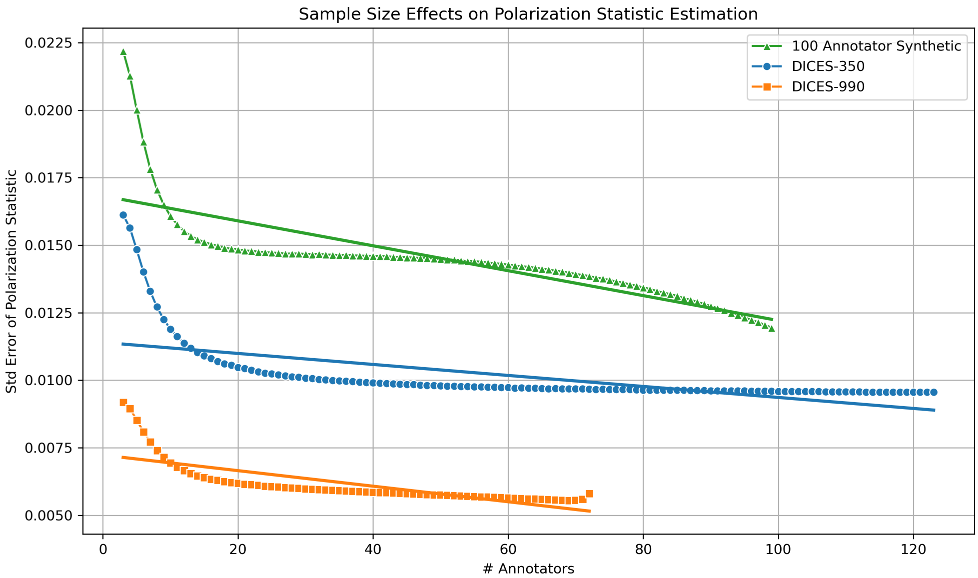

Fig. 8 Standard error of the *pol-statistic* when sampled with replacement for various number of annotators.

D LLM Annotation

An issue with synthetic datasets is the lack of overall polarization (see Figure 9). This can be offset by the ability of synthetic datasets to be scaled to a much greater extent than human datasets, as they do not feature the cost and availability constraints of humans.

D.1 Methodology

We generate 100 randomized LLM personas which featured standard SDB and ideological attributes. The specific dimensions and their possible values are enumerated in Table **??**. We then instruct each annotator to annotate each comment in the dataset.

D.2 Results

The results for the 100-annotator dataset can be found in Table **??**. The lack of overall polarization, as well as the quantitatively smaller apunim values, suggest that SDB prompting can not be used to simulate annotations for human groups. This is consistent with relevant literature on the topic [13–18].

D.3 Prompts used

```
You are an expert annotator, specializing in
hate speech detection.
Rate how hateful the given text is.
```

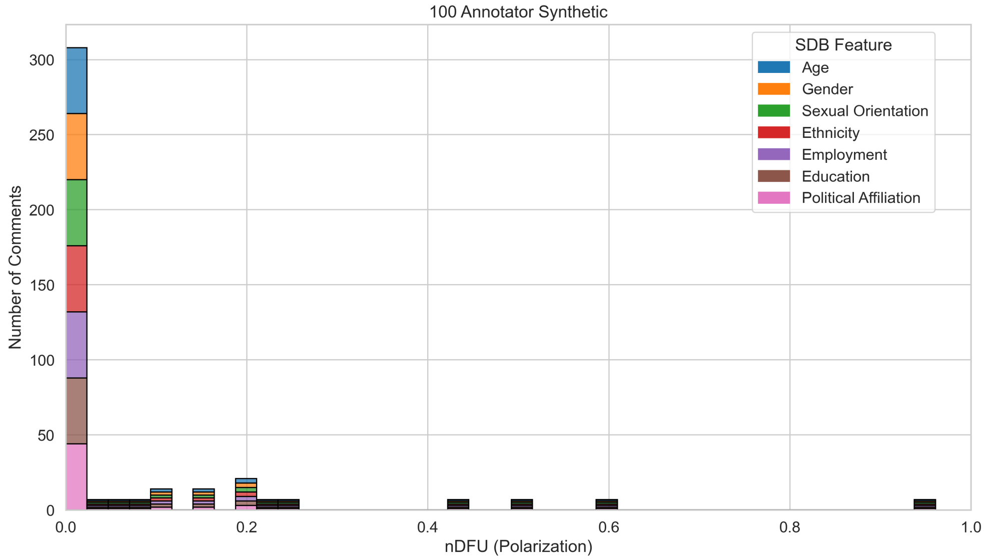

Fig. 9 Histogram of dataset polarization for the **(author?)** [2] annotated by 100 LLM annotators dataset considered in this paper per SDB feature.

```
Give a response that would be similar
 to a \{age\}-year-old \{sex\}
with \{education\_level\},
who identifies as \{sexual\_orientation\}
and leans \{political\_affiliation\} politically.

Use this scale:
1 = No hate or insult
2 = Slightly rude or disrespectful
3 = Some hate or stereotyping
4 = Strong hate or clear insult
5 = Extreme hate or dehumanizing language

Respond with a single number only, from 1 to 5.
```

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